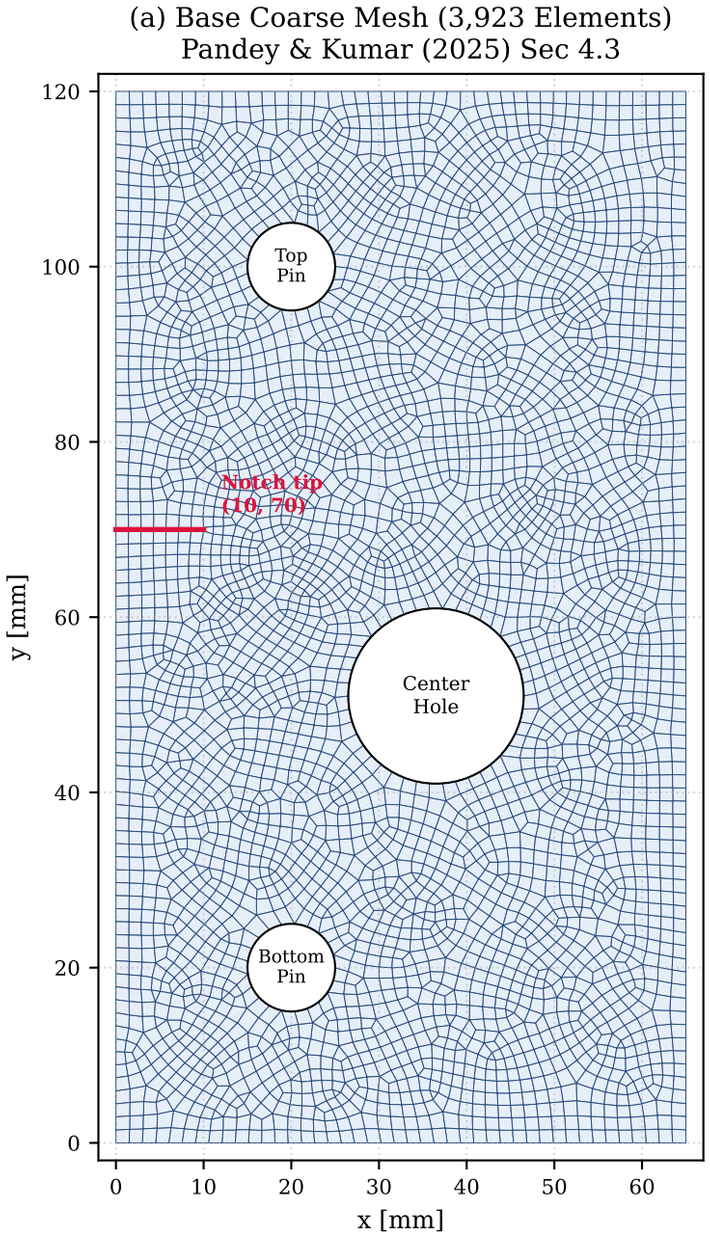
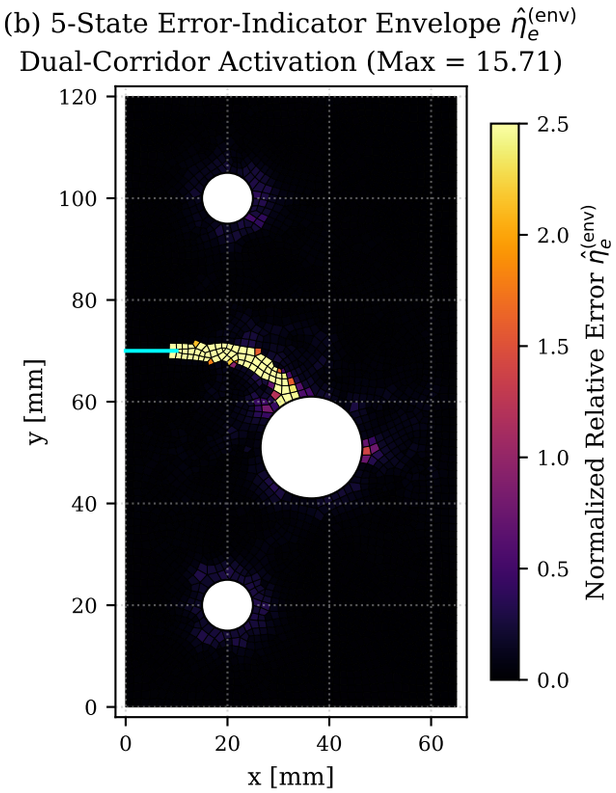
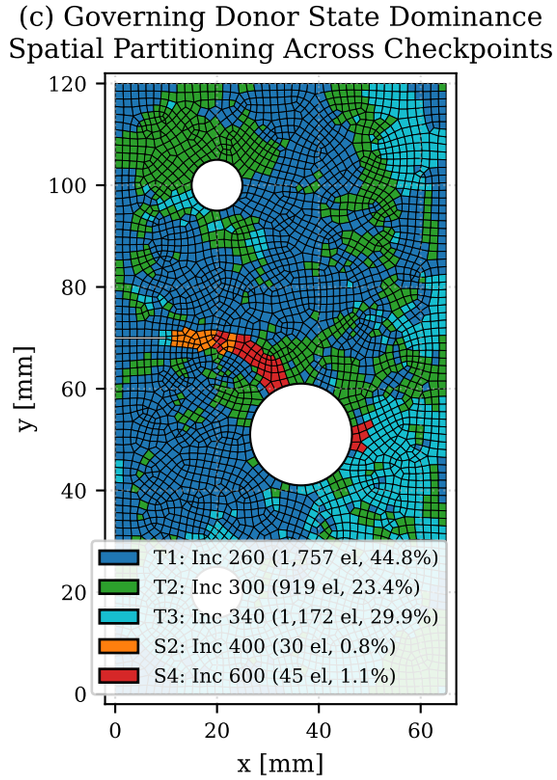
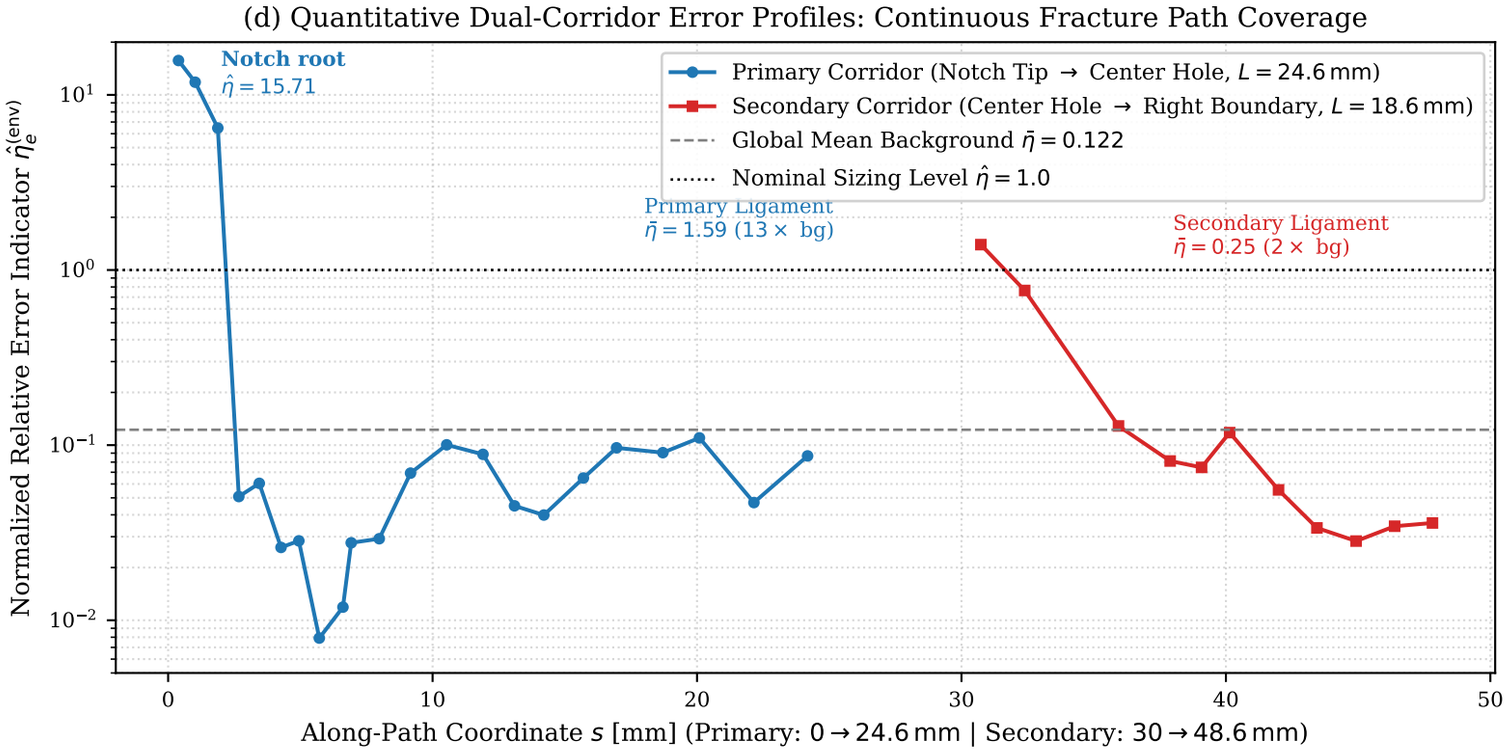

(a) Base Coarse Mesh (3,923 Elements)
Pandey & Kumar (2025) Sec 4.3
Top Pin
Center Hole
Bottom Pin
Notch tip (10, 70)
x [mm]
y [mm]


(b) 5-State Error-Indicator Envelope $\hat{\eta}_e^{(env)}$
Dual-Corridor Activation (Max = 15.71)
Normalized Relative Error $\hat{\eta}_e^{(env)}$
x [mm]
y [mm]


(c) Governing Donor State Dominance
Spatial Partitioning Across Checkpoints
T1: Inc 260 (1,757 el, 44.8%)
T2: Inc 300 (919 el, 23.4%)
T3: Inc 340 (1,172 el, 29.9%)
S2: Inc 400 (30 el, 0.8%)
S4: Inc 600 (45 el, 1.1%)
x [mm]
y [mm]


(d) Quantitative Dual-Corridor Error Profiles: Continuous Fracture Path Coverage
Notch root
$\hat{\eta} = 15.71$
Primary Corridor (Notch Tip → Center Hole, $L = 24.6$ mm)
Secondary Corridor (Center Hole → Right Boundary, $L = 18.6$ mm)
Global Mean Background $\bar{\eta} = 0.122$
Nominal Sizing Level $\hat{\eta} = 1.0$
Primary Ligament
$\bar{\eta} = 1.59$ (13 × bg)
Secondary Ligament
$\bar{\eta} = 0.25$ (2 × bg)
Normalized Relative Error Indicator $\hat{\eta}_e^{(env)}$
Along-Path Coordinate s [mm] (Primary: 0 → 24.6 mm | Secondary: 30 → 48.6 mm)